Implementation of Potential Leukemia Detection using Recurrent Neural Networks with Blood Cell Counting

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***Abstract*—A total of 412,000 persons are expected to be diagnosed with leukemia worldwide, with acute lymphoblastic leukemia accounting for around 12% of all cases. Thus, the early detection of leukemia can save millions of populations. This article mainly focusing on blood cell counting and detection of Leukemia using deep learning mechanisms. Initially, the images are preprocessed using median filters and segmented using RGB-Otsu's approach. Then, the features are extracted using scale-invariant feature transform (SIFT) and these features are applied to recurrent neural network (RNN) for classification. Finally, the proposed method resulted in superior performance as compared to the state of art machine learning approaches.**

Keywords—Scale-invariant feature transform, Recurrent neural network, Deep learning and RGB-OTSU.

I. INTRODUCTION

Computerized cancer detection and therapy is one of the hottest areas in deep learning. The discovery of cancer markers, a sketch of cancer growth activity, and the staging of distinct cancer types are all usually included. Several convolutional neural network-based deep learning models with excellent performance and accurate identification have the potential to enhance medical tasks. In some diseases, breakthroughs and improvements in computer algorithms and hardware devices, as well as clever algorithms applied to medical imagery, provide diagnosis accuracy that clinicians cannot match. This research examines the evolution of cancer detection from classic computer-aided approaches to convolutional neural networks, and shows how practical use of convolutional neural networks may change the detection model from laboratory to clinical application using real-world examples. Cancers are abnormal growths of tissue that are damaging to one's health.

The malignant cancer is one of the most common illnesses that has a negative impact on human health and poses a threat to human life. To successfully minimize cancer mortality, early diagnosis of abnormal characteristics is critical for cancer treatment. Routine laboratory examinations of the patient's secretions, as well as serum, immunological, and genetic studies, are all traditional diagnostic approaches. X-ray, CT, MRI, SPECT scan, and other clinical imaging studies are now in use. With the introduction of new difficulties relating to radiation noise reduction, researchers are increasingly looking into medical Blood Smear noise reduction technologies. At the same time, in the follow-up diagnosis of lesions, clinicians frequently depend on clinical experience and academic background knowledge. Promoting clinical diagnostic technology, on the other hand, is difficult. As a result of the medical demands, medical imaging technology and computer-aided diagnosis research emerges. The benefits of using a convolutional neural network to diagnose cancers are becoming more apparent. In the business, research on computer-aided diagnosis based on medical imaging of malignancies has gotten a lot of attention.

II. LITERATURE SURVEY

In [6] authors used the machine learning approach is used to address this issue and improve performance. When the source domain and the destination domain are wholly distinct, transfer learning has no effect on performance on medical blood smear tasks. It has the potential to decrease performance in some circumstance. The classic approaches 7[] of computer-aided cancer diagnostics are discussed in this work. It covers cancer Blood Smear segmentation and classification, as well as CNN-based diagnosis approaches for assisting clinicians in determining cancers. It serves as a model for the development of a CNN computer-aided system based on cancer detection research. As a result, unlike previous approaches that employed the ImageNet dataset for machine learning, this work used the same domain transfer learning.

In [8] authors introduced the conversion of histological data slides to digital blood Smears utilizing advanced computational technologies to simplify the collecting, assessment, storage, and transportation of histologic information is known as digital pathology. Analog histology data may be analyzed using deep learning/artificial intelligence (DL/AI) approaches [9] because to the nature of digitalization.

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In [10] authors used cases and regulatory framework for deploying such systems in the clinical environment are still in the early stages, the application of DL/AI to digital pathology data has promise. Recent research [11] employing whole-slide photos and deep learning/artificial intelligence to detect histologic abnormalities in general, and cancer in particular, have yielded promising results. In [12] authors work will focus on these developing technologies in diagnostic hematology and the assessment of lymphoproliferative disorders in this study. As a result, detecting normal and cancerous cells in a reliable and precise manner is crucial.

In [13] authors used many data augmentation strategies to reduce the overfitting impact. The transfer learning approach is used to address this issue and improve performance [14]. when the source domain and the destination domain are wholly distinct, transfer learning has no effect on performance on medical blood smear tasks. It has the potential to decrease performance in some circumstances [15]. As a result, unlike previous approaches that employed the ImageNet dataset for transfer learning, we used the same domain transfer learning. This work used many data augmentation strategies to reduce the overfitting impact.

III. Proposed Methodology

Blood smear processing technology has advanced quickly in recent years and is becoming increasingly frequently employed. Some of these technologies are well-established and have shown to be beneficial. Image processing's major aim is to develop a greater range of applications by researching novel processing methods. High precision, rich information processing, flexible processing, and sophisticated non-linear processing are all advantages of traditional digital image processing. The drawback is that for sophisticated processing, the processing speed is sluggish. The spatial and transform domain approaches of digital image processing may be classified into two groups. This article gives the detailed analysis of blood cell counting followed by Leukemia detection classification. Figure 1 presents the detailed operation of proposed method.

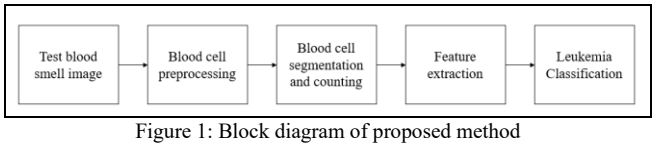


Figure 1: Block diagram of proposed method

A. Preprocessing

Image pre-processing, segmentation, feature extraction, feature selection, and classification are all interrelated phases in proposed technique. Blood Smear enhancements such as contrast correction, noise reduction, and image sharpening are included in the pre-processing stage. Before image segmentation, all of these stages are applied to the input Blood Smear. The median filter, low pass filter, high pass filter, and Gabor filter are all examples of pre-processing methods. Before Blood Smear segmentation, they are commonly employed for image contrast correction, image sharpening, and noise reduction. Several studies have looked at proposed for the detection and classification of leucocytes. However, employing these methods, effective nuclei identification, border separation to recover overlapping cells, ROI segmentation, robust feature extraction, and optimal feature selection have become difficult and time-consuming. Following ROI segmentation, feature extraction is the next stage in this method. The classification in classic supervised learning approaches is dependent on selecting a robust features descriptor and the optimal features selection algorithm, which are the most important elements in ensuring the efficiency and accuracy of the methodology.

B. Blood cell counting and segmentation

In computer-assisted diagnostic systems, blood smear segmentation is critical. Its goal is to separate feature areas from the remainder of the image. To distinguish between distinct sections of an image, it can integrate visual elements such as texture information with other biological aspects. As a result, unaffected areas may be automatically filtered out, while suspect areas with pathological characteristics like uneven texture can be left behind. For treatment planning, determining the cancer's location and size, as well as correct segmentation data, is critical. Image segmentation is a very effective strategy to identify cancer among the various medical imaging technologies.

However, due to variances in biological information in various parts of the human anatomy, human involvement is usually always necessary. An expert doctor must supply the basic settings for setting categorization or training data. However, multiple researches have been conducted to identify various forms of cancer using data derived from medical imaging. Most image-based cancer detection techniques, on the other hand, only identify a single cancer mound. The RGB-OTSU technique is a basic classification and regression approach. From wide-field microscopy blood smears pixel images of individual cells. The proposed RGB-OTSU algorithm first recognized individual cells by utilizing an Otsu threshold on the DAPI channel to detect cell nuclei as shown in Figure 2.

Small blood smear patches centered on each of the nuclei were clipped out and sent through a series of tests to exclude images with numerous cells, nuclei, dead cells, and debris. First, Blood Smear patches containing a cell with insufficient were discarded as non-viable. Second, the watershed method was used to each potential image patch to see if numerous items were present in the nucleus or cytoplasm channels; if multiple objects were discovered, the image patch was eliminated since only single-cell blood smears were counted. Finally, any Blood Smear patches with distinct fluorescently stained detritus, perhaps from dead cells, were discarded. Finally, each blood smear was counted for the presence of the three fluorescent stains using a minimum fluorescence emission threshold. This digital processing resulted in a series of high-quality single-cell photographs that were stained appropriately with each of the three fluorescent dyes.

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C. Feature extraction

The SIFT operator employs 128-dimensional feature vectors to describe the feature points observed in the image, the image following the SIFT technique is represented as a set of 128-dimensional feature vectors. The feature vector set is an outstanding local feature description approach that is invariant to Blood Smear scaling and translation. The SIFT approach involves pole identification in scale space, precise localization of key points, key point orientation determination, and feature vector production. The location and scale of feature points must be known for feature point identification, and the feature points to seek are those whose position does not vary in continuous scale space. Finally, the Gaussian pyramid (GP) is illustrated in Figure 3, is created using a series of scale-space modifications and second sub-sampling. Through the GP, the scale of space is realized.

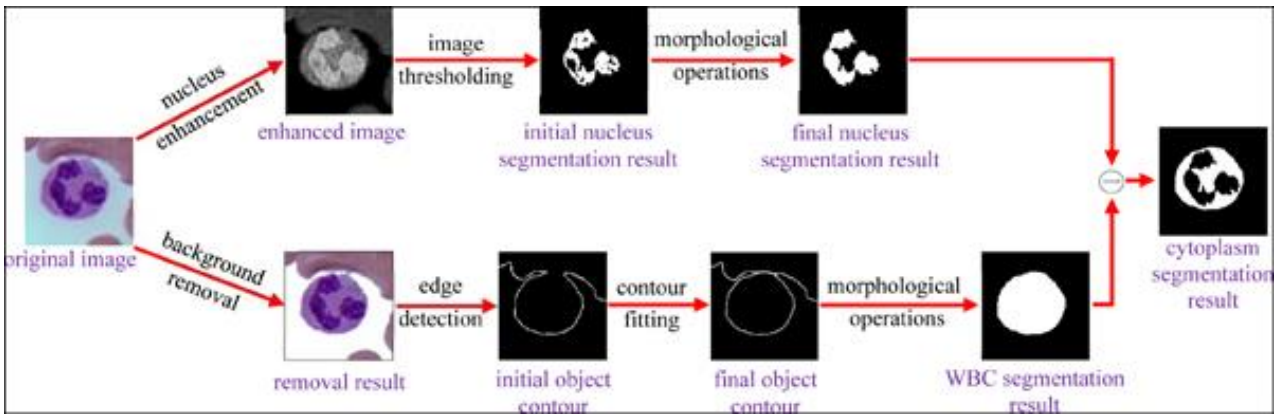


Figure 2. RGB-OTSU segmentation

The pyramid model of a photograph refers to a tower-like model that gradually diminishes the order of the original image to produce a set of blood smears of various sizes, from tiny to huge and from top to bottom. Each pyramid has n tiers in total. The initial image is the first layer of the golden tower, and each downward sampling produces a new image of a layer of the pyramid. The number of levels in the pyramid is determined by the size of the top Blood Smear and the original image.

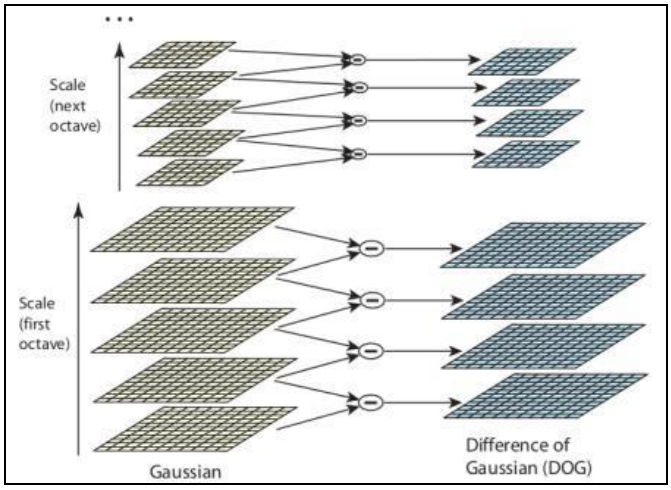


Figure 3. SIFT feature extraction

A Gaussian filter is applied to the Gaussian pyramid based on simple down sampling. Each layer of the pyramid has several Gaussian blur Blood Smears, which are referred to as octaves in the pyramid. In extremum detection, using the difference of Gaussians (DOG) pyramid instead of the Gaussian pyramid might save a lot of time. Finding the extreme point in DOG, which is the candidate point of the feature point, is the next stage in creating the differential GP. The key point's candidate point is the extremum point, which is made up of the DOG space local extremum points. By comparing two adjacent layers of photos of each DOG in a group, the critical spots were first discovered. In order to find the DOG function extreme point, each pixel must be compared to all of its neighboring points in the scale and image domain, regardless of whether they are smaller or bigger.

D. RNN Classification

This article implemented the RNN in depth, modelling the zones created by detection-based systems to determine which is the best in this situation when doing two tasks: blood cell counting and leukemia classification. Further, RNNs outperform typical machine learning algorithms mixed with hand-crafted descriptors in such a circumstance, according to experimental data. Even basic techniques can yield to reliable results in both tasks when using acceptable photos for model training. Leukemia is a disorder that damages the bone marrow's ability to produce white blood cells. Cancer cells are created in an improper manner, displacing regular blood cells. As a result, the individual has difficulty transferring oxygen and battling infections. Because of the high degree of accuracy required and the high danger of minute mistakes, applying machine learning techniques to the medical sector is particularly difficult.

The application of these approaches to a more sophisticated subdomain of hematological diagnosis, such as automated identification of blood cell types, which can aid in the detection of hematopoietic illnesses, appears to be extremely promising. On the microscopic peripheral blood cell dataset, the RNN compare 27 prominent deep features. The dataset is open to the public, and it contains a significant number of normal peripheral blood cells.

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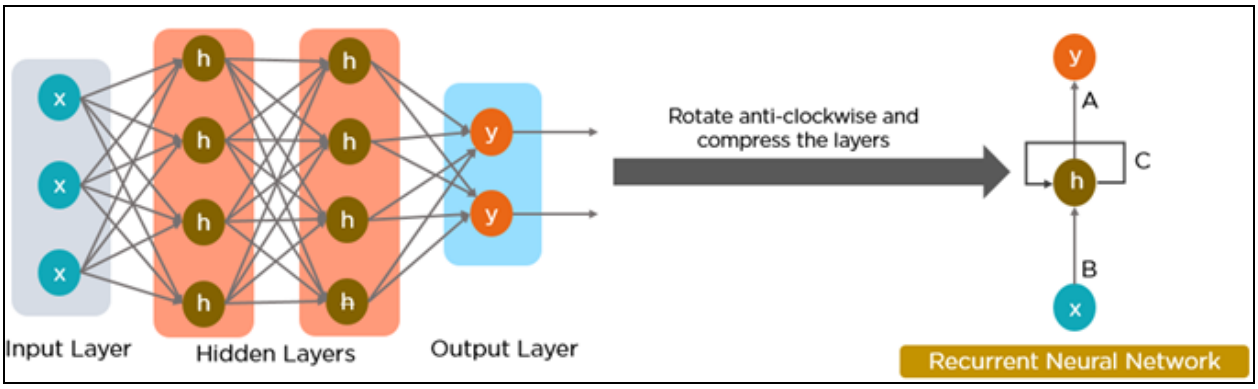


Figure 4. RNN classification

For leukemia classification, RNN fine-tune state-of-the-art image classification models pre-trained on the ImageNet dataset. To minimize overfitting and improve generalization, RNN use data augmentation approaches during training. Thus, the utilization of RNNs in the investigation of intelligent approaches to aid medical blood smear diagnosis has made great progress through multiple number of hidden layers.

IV. Simulation Results

The simulations are carried out using the MATLAB programming language, and the MATLAB R2018a tool is used for both segmentation and categorization operations.

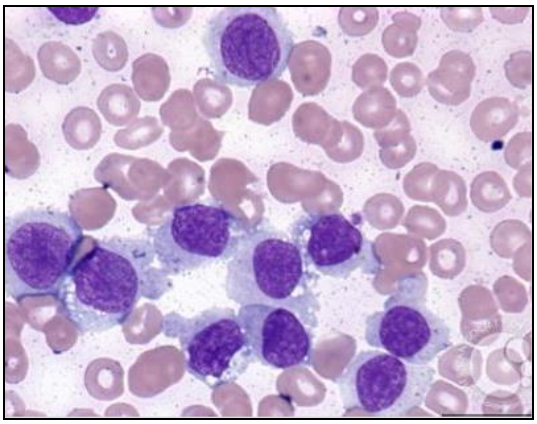


Figure 5. Input image

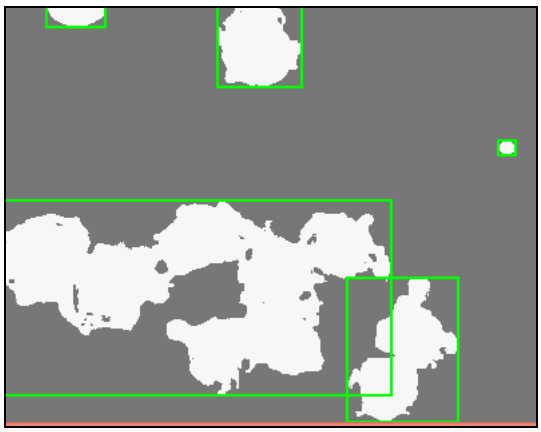


Figure 6. White Blood cell segmented image

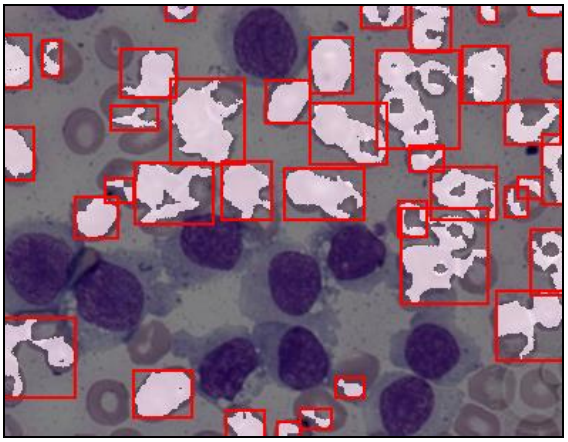


Figure 7. Red blood cell segmented image

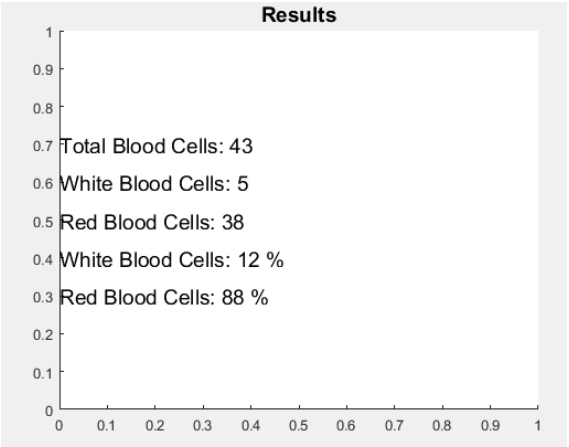


Figure 8. Blood cell count outcomes

Figure 5 presents the pre-processed input image with white blood cells are segmented in figure 6 and figure 7 represents the red blood cell segmented image, respectively. Finally, the blood cell counted results are presented in the figure 8, respectively.

Table I. Performance comparison

Method	Accuracy	Sensitivity	Specificity	F1-score
SVM [8]	0.7157	0.78099	0.85796	0.47765
k-means [9]	0.77985	0.90715	0.99999	0.99999
ANN [11]	0.80588	0.90024	0.9166	0.83857
Proposed	0.98931	0.96198	1	1

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Table I shows that, the performance of proposed RNN based leukemia cancer detection performed in superior when compared to the various kernels of SVM [8], k-means [9] and ANN [11] for all metrics.

V. Conclusion

A total of 412,000 people is predicted to be diagnosed with leukemia across the world, with acute lymphoblastic leukemia accounting for around 12% of cases. As a result, early identification of leukemia can save the lives of millions of people. This article focuses on applying deep learning processes to count blood cells and identify leukemia. The photos are first preprocessed using median filters and then segmented using RGB-method. Otsu's segmented features are then retrieved using the SIFT, and these features are used to classify the data using an RNN. Finallsy, when compared to state-of-the-art machine learning algorithms, the proposed strategy outperformed them. This work can be extended to implement with deep learning CNN classification with various blood cancer types on real time datasets.

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